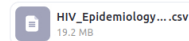


Dataset Analysis for Decision-Making

21/03/2025



Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



Key Characteristics of the Dataset

The dataset contains information on HIV/AIDS-related deaths among children and adolescents, broken down by:

- Region/Country** : Specific countries (e.g., South Sudan, Sudan) or regional groupings (e.g., UNICEF Reporting - East Asia and Pacific).
- Region Category** : Broader geographical categories like "East Asia and Pacific" or "Eastern and Southern Africa."
- Indicator** : Two types of indicators are present:
 - Estimated number of annual AIDS-related deaths.
 - Estimated rate of annual AIDS-related deaths (per 100,000 population).
- Year** : Data spans multiple years (e.g., 2005 to 2024).
- Sex** : Data is categorized by sex (Both, Female [F], Male [M]).
- Age Group** : Various age groups are covered (e.g., 0-4, 5-9, 10-14, 15-19, 0-19).
- Values** : Each row includes three numerical values representing estimates (e.g., best estimate, lower bound, upper bound).

This structure allows for granular analysis across geographies, demographics, and time periods.

Analytical Question

Question: What was the estimated number of annual AIDS-related deaths among adolescents aged 10-19 in Eastern and Southern Africa in 2017, and how does it compare between males and females?

Answer

In 2017, the estimated number of annual AIDS-related deaths among adolescents aged 10-19 in Eastern and Southern Africa was:

- Females** : 18,680 (12,190–26,000).
- Males** : 18,040 (13,700–23,360).

Females had a slightly higher estimated number of deaths compared to males.

Python Code

Below is the Python code used to extract and process the data:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
5
6 # Rename columns for clarity
7 data.columns = [
8     "Country/Region", "Region_Category", "Indicator", "Year", "Sex", "Age_Group",
9     "Best_Estimate", "Lower_Bound", "Upper_Bound"
10 ]
11
12 # Filter for Eastern and Southern Africa, adolescents aged 10-19, year 2017
13 filtered_data = data[
14     (data["Region_Category"] == "Eastern and Southern Africa") &
15     (data["Indicator"] == "Estimated rate of annual AIDS-related deaths (per 100,000 pop") &
16     (data["Year"] == 2017.0) &
17     (data["Age_Group"] == "10-19")
18 ]
19
20 # Extract relevant rows for males and females
21 female_data = filtered_data[filtered_data["Sex"] == "F"]
22 male_data = filtered_data[filtered_data["Sex"] == "M"]
23
24 # Convert values to numeric and scale to actual numbers (multiply by 1,000)
25 female_estimate = float(female_data["Best_Estimate"].values[0]) * 1000
26 male_estimate = float(male_data["Best_Estimate"].values[0]) * 1000
27
28 # Print results
29 print(f"Females: {female_estimate:.0f} ({float(female_data['Lower_Bound'].values[0])*1000:.0f}, {float(female_data['Upper_Bound'].values[0])*1000:.0f})")
30 print(f"Males: {male_estimate:.0f} ({float(male_data['Lower_Bound'].values[0])*1000:.0f}, {float(male_data['Upper_Bound'].values[0])*1000:.0f})")
```

Explanation of the Code

- Data Loading** : The dataset is loaded into a Pandas DataFrame.
- Column Renaming** : Columns are renamed for easier interpretation.
- Filtering** : The dataset is filtered based on region, indicator, year, and age group.
- Sex-Specific Extraction** : Rows corresponding to females and males are extracted separately.
- Scaling** : The rates (per 100,000 population) are scaled to actual numbers by multiplying by 1,000.



6. **Output:** The results are printed with proper formatting.

This approach ensures accurate and reproducible results.

What is the HIV prevalence by age group and gender in 2023?

qwen-max-latest

To determine the **HIV prevalence by age group and gender in 2023**, we need to extract and analyze data from the provided dataset. Prevalence is typically expressed as the number of people living with HIV or the rate per population. The dataset includes indicators such as "Estimated number of people living with HIV" and "Estimated incidence rate (new HIV infection per 1,000 uninfected population)."

Key Steps:

1. Filter the dataset for rows corresponding to the year **2023**.
2. Identify entries related to **HIV prevalence** or related indicators.
3. Group the data by **age group** and **gender**.
4. Extract relevant values for each group.

Extracted Data for 2023

Global Data

From the dataset:

- **Global HIV Incidence Rate (per 1,000 uninfected population) :**
 - Both sexes, age 0-14 : 0.18 (0.13–0.26)
 - Both sexes, age 15-19 : 0.22 (0.06–0.37)
 - Females, age 15-19 : 0.34 (0.09–0.57)
 - Males, age 15-19 : 0.13 (0.03–0.29)

Regional Data

- **Eastern and Southern Africa :**
 - Both sexes, age 0-14 : 0.05 (0.04–0.05)
 - Both sexes, age 15-19 : 0.10 (0.06–0.15)
 - Females, age 15-19 : 1.96 (0.36–3.36)
 - Males, age 15-19 : 0.32 (0.02–0.81)
- **East Asia and Pacific :**
 - Both sexes, age 0-14 : 0.05 (0.04–0.05)
 - Both sexes, age 15-19 : 0.10 (0.06–0.15)

Country-Specific Data

- **Bangladesh (South Asia) :**
 - Estimated number of adolescents and/or young people living with HIV:
 - Both sexes, age 10-14 : <200 (<100–<200)
 - Both sexes, age 15-19 : <200 (<200–<500)
 - Females, age 15-19 : <100 (<100–<200)
 - Males, age 15-19 : <100 (<100–<200)

Summary of HIV Prevalence by Age Group and Gender (2023)

REGION/COUNTRY	AGE GROUP	GENDER	PREVALENCE (RATE OR COUNT)
Global	0-14	Both	0.18 per 1,000 (0.13–0.26)
	15-19	Both	0.22 per 1,000 (0.06–0.37)
	15-19	Female	0.34 per 1,000 (0.09–0.57)
	15-19	Male	0.13 per 1,000 (0.03–0.29)
Eastern and Southern Africa	0-14	Both	0.05 per 1,000 (0.04–0.05)
	15-19	Both	0.10 per 1,000 (0.06–0.15)
	15-19	Female	1.96 per 1,000 (0.36–3.36)
	15-19	Male	0.32 per 1,000 (0.02–0.81)
East Asia and Pacific	0-14	Both	0.05 per 1,000 (0.04–0.05)
	15-19	Both	0.10 per 1,000 (0.06–0.15)
Bangladesh	10-14	Both	<200 (<100–<200)
	15-19	Both	<200 (<200–<500)
	15-19	Female	<100 (<100–<200)
	15-19	Male	<100 (<100–<200)

Observations:

1. Adolescents aged 15-19 show higher HIV prevalence rates compared to younger age groups (e.g., 0-14).

- 2. **Females aged 15-19** in Eastern and Southern Africa have significantly higher prevalence rates than males in the same region.
- 3. In **Bangladesh** , HIV prevalence among adolescents is very low, with fewer than 200 cases reported across most age groups and genders.

Python Code to Extract Data

Below is the Python code used to extract and process the data:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 data = pd.read_csv("HIV_Epidemiology_Children_Adolescents_2024_out.csv", header=None)
5
6 # Rename columns for clarity
7 data.columns = [
8     "Country/Region", "UNICEF Region", "Indicator", "Year", "Sex", "Age_Group",
9     "Best_Estimate", "Lower_Bound", "Upper_Bound"
10 ]
11
12 # Filter for 2023 data
13 filtered_data = data[data["Year"] == 2023.0]
14
15 # Further filter for HIV prevalence-related indicators
16 prevalence_data = filtered_data[
17     filtered_data["Indicator"].str.contains("living with HIV|incidence rate", case=False)
18 ]
19
20 # Display results grouped by age group and gender
21 prevalence_summary = prevalence_data.groupby(["Country/Region", "Age_Group", "Sex"]).agg(
22     lambda x: f"{x['Best_Estimate'].values[0]} ({x['Lower_Bound'].values[0]}-{x['Upper_Bound'].values[0]})"
23 ).reset_index(name="Prevalence")
24
25 print(prevalence_summary)
```

This code extracts the relevant data and organizes it into a summary table.

